

test3 magnetism, source of magnetism

 This is a preview of the published version of the quiz

Started: Apr 7 at 12:26pm

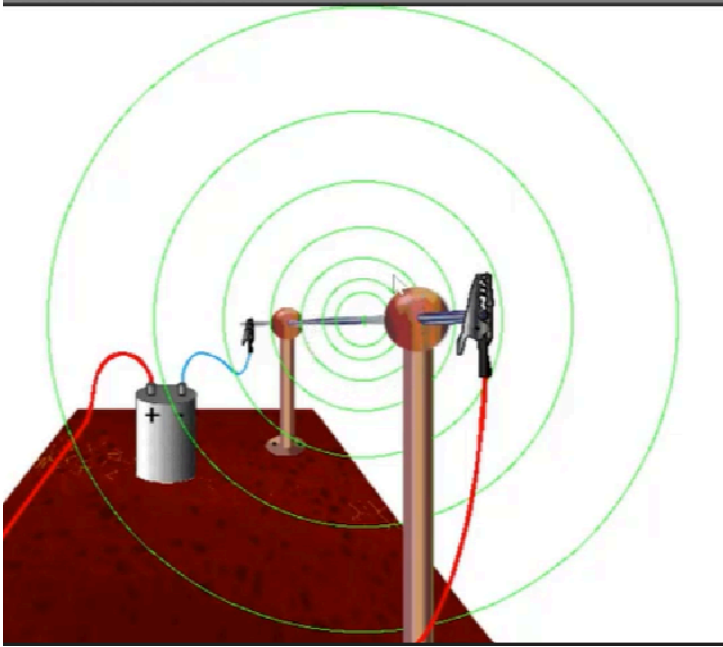
Quiz Instructions

It will close at 11:59PM. You can not see your answers. correct or not correct. I can not reopen the test. So be careful.

It is open note, open discussion, open notebook, open brain. good luck



Question 1 6 pts



A solid wire is connected to a power supply (12VDC). You can see the terminals of the battery. You close the switch and current flow through the solid wire. The green circles are the magnetic field lines. The field

- ☐ circles CW
- ☐ oscillates CW then CCW then CW ..
- ☐ Can't tell
- ☐ circles CCW



Question 2 6 pts

Building up on the previous question. If the magnetic field is 10 Gauss at a distance 10cm from the conductor, the magnetic field at a distance 5cm is



5 Gauss



10 Gauss



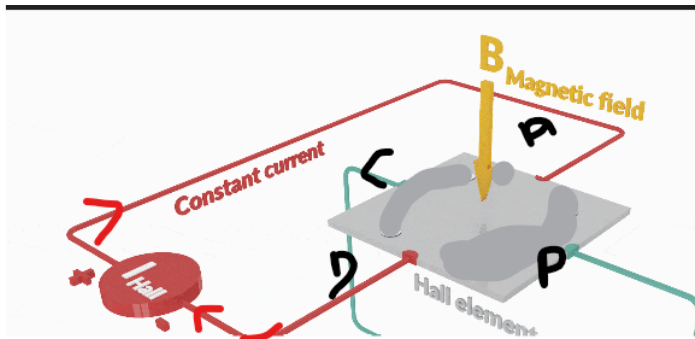
20 Gauss



40 Gauss



Question 3 6 pts



A current (red) flows from the battery (red +) through a conductive plate (grey) back to the battery. A magnetic field is introduced as shown (the direction is down). The positive charge (current) will be deflected toward



P



C



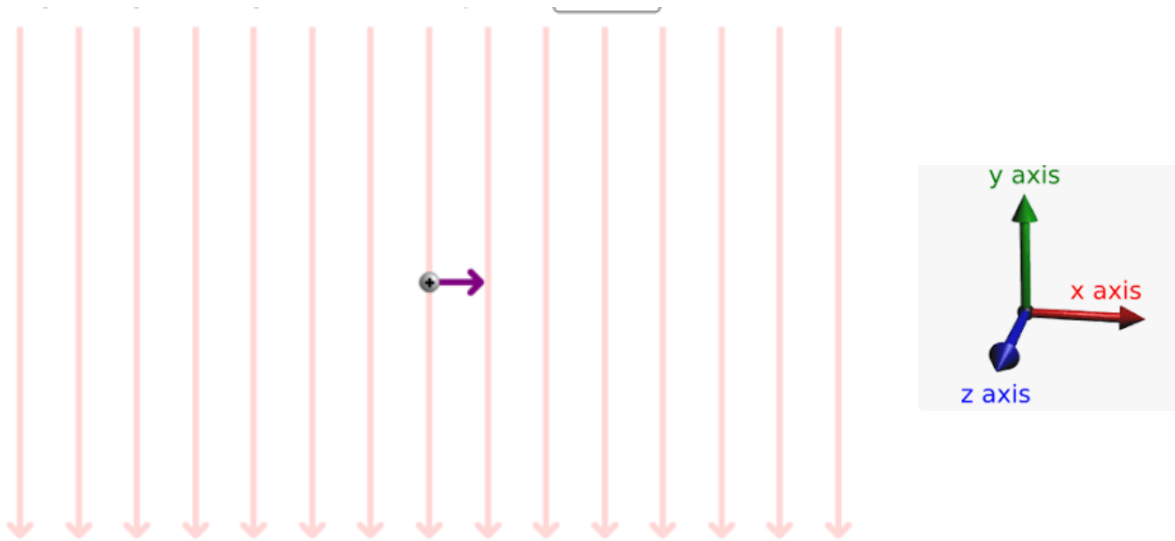
will not be deflected



A



Question 4 6 pts



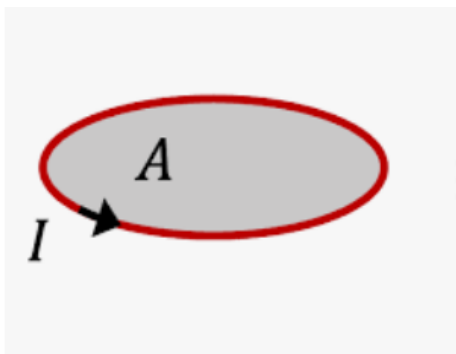
at $t=0$ A positive charge has a velocity V_x (shown in purple). The initial velocity is along the positive x -axis (right). an electric field is applied. Shown in pink. The electric field is along the negative y -axis (down).

The charge

- ☐ the charge has no acceleration and its motion is a line along the positive x -axis. The velocity is constant.
- ☐ will accelerate up ($a_y > 0$) while moving at a constant speed (V_x). It describes a parabola curved up
- ☐ will accelerate down ($a_y < 0$) while moving at a constant speed (V_x). It describes a parabola curved down.
- ☐ The charge will describe a spiral increasing in size. It will speed up while going into a circle.
- ☐ the charge will describe a uniform circular motion. It has a centripetal acceleration. The magnitude of the velocity stays the same (V_x)



Question 5 5.5 pts



This loop of current is perpendicular to the plane of your screen. It is

- ☐ Is equivalent to a North monopole



equivalent of a bar magnet. The South is above area A



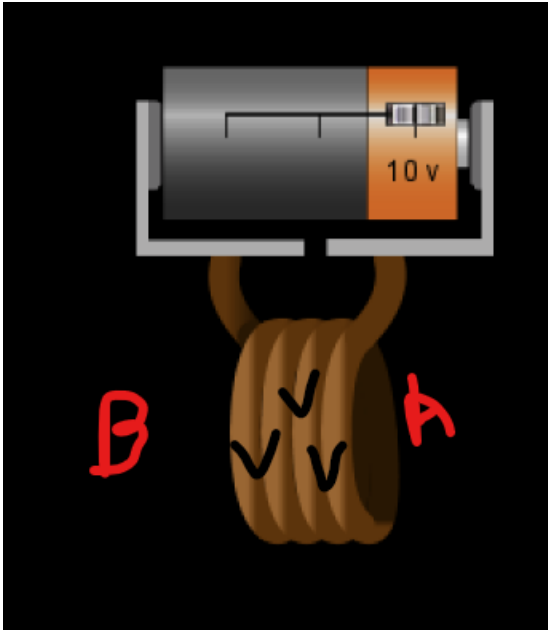
is equivalent of a South monopole



equivalent to a bar magnet. The North is above the area A



Question 6 5.5 pts



I have drawn the direction of current. What is true



the set up is a North monopole



Face A is the North



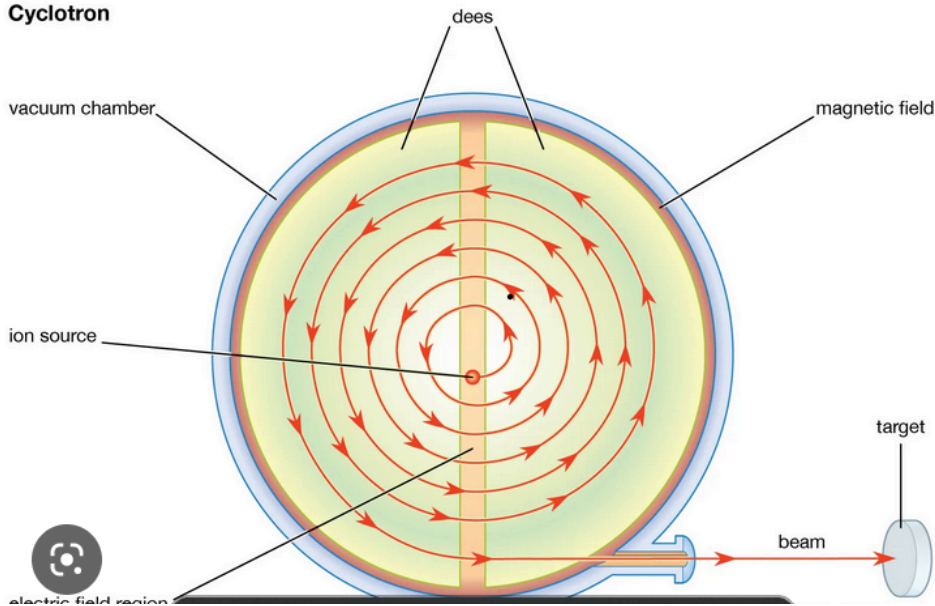
Face B is the North



the set up is a south monopole



Question 7 5.5 pts

Cyclotron

A positive particle is accelerated in a cyclotron. The uniform magnetic field is

☐

@ right

☐

in your screen (away from you)

☐

out of your screen (toward you)

☐

@ left

⋮

Question 8 5.5 pts

A very long solenoid (use the equation from the slides). has a length of 500cm and a radius of 1 cm. It has 100 loops of current. Its magnetic field is 1T. If the radius is now twice as large, then

hint: scrutinize the equation for solenoid

☐

the magnetic field is 2T

☐

the magnetic field is 4T

☐

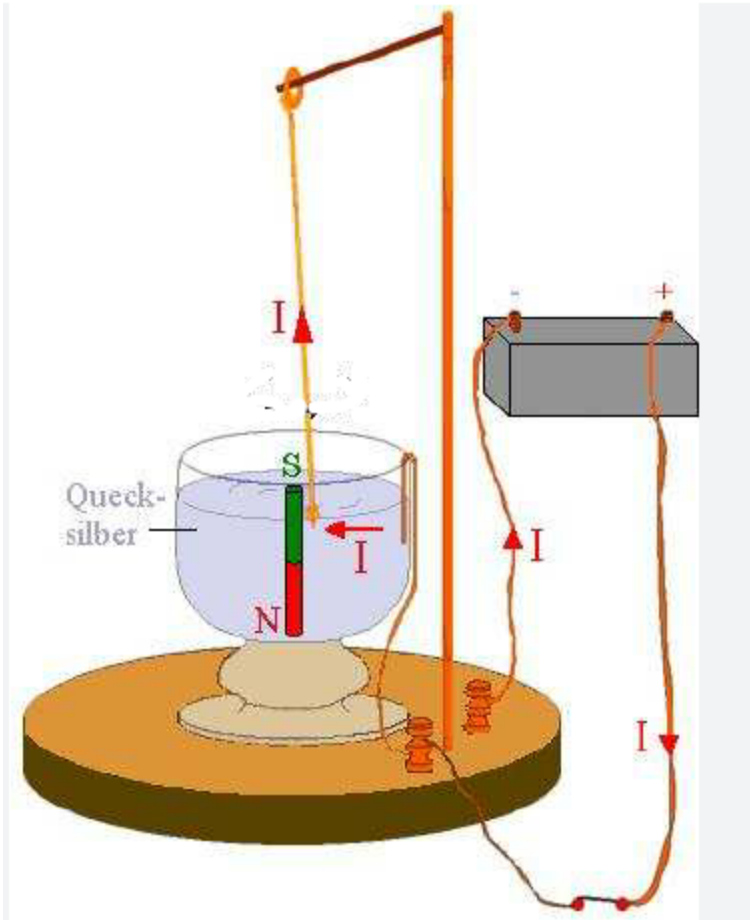
the magnetic field is 0.5T

☐

the magnetic field is 1T

⋮

Question 9 5.5 pts



Here is a diagram of the first motor built by Faraday. The yellow solid wire is mobile and will start spinning around the magnet. The liquid is mercury so its a conductor. In the situation descibed above, the magnetic force acting on the yellow wire inside the liquid is: (given the magnetic field produced by the magnet and direction of I . the red arrow @ right)

- ☐ the force is in the screen (away from you)
- ☐ the force is out of the screen. Toward you.
- ☐ there is no force
- ☐ the force is @ right
- ☐ the force is @ left.



Question 10 5.5 pts

A proton, which moves perpendicular to a magnetic field of 1.2 T in a circular path of radius 0.080 m, has what speed? ($q_p = 1.6 \times 10^{-19} \text{ C}$ and $m_p = 1.67 \times 10^{-27} \text{ kg}$)

a. $3.4 \times 10^6 \text{ m/s}$

b. $4.6 \times 10^6 \text{ m/s}$

c. $9.6 \times 10^6 \text{ m/s}$

d. $9.2 \times 10^6 \text{ m/s}$

☐

d

☐

b

☐

a

☐

c

☒

Question 11 5.5 pts

A copper wire of length 25 cm is in a magnetic field of 0.20 T. If it has a mass of 10 g, what minimum current through the wire would cause a magnetic force equal to its weight?

a. 1.3 A

c. 2.0 A

b. 1.5 A

d. 4.9 A

Round to the nearest tenth. convert to kg

☐

A

☐

C

☐

D

☐

B

☒

Question 12 5.5 pts

A circular loop carrying a current of 1.0 A is oriented in a magnetic field of 0.35 T. The loop has an area of 0.24 m^2 and is mounted on an axis, perpendicular to the magnetic field, which allows the loop to rotate. If the plane of the loop is oriented parallel to the field, what torque is created by the interaction of the loop current and the field?

a. 5.8 N·m

c. 0.084 N·m

b. 0.68 N·m

d. 0.017 N·m

☐

A

☐

B

☐

D

☐

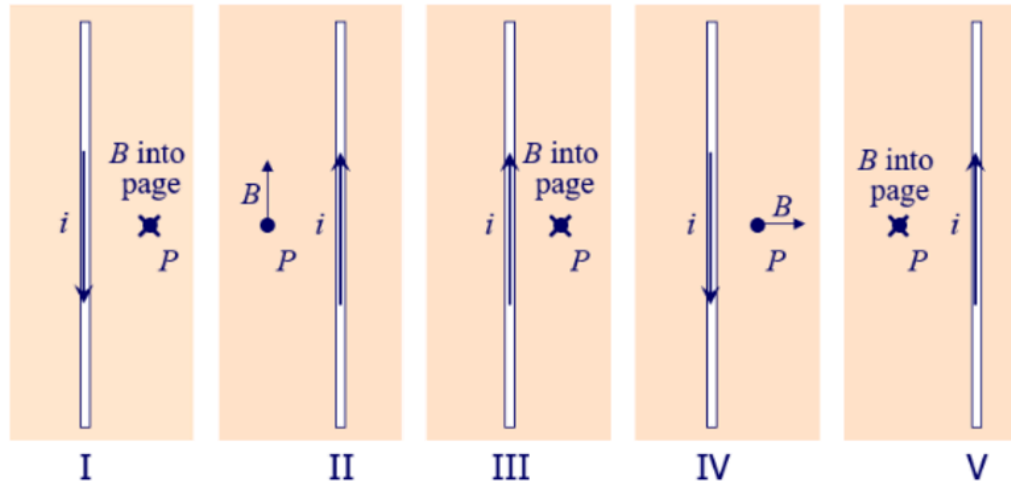
C

⋮

Question 13 5.5 pts

1. Which drawing below shows the correct direction of the magnetic field, B , at the point P ?

- A. I.
 B. II.
 C. III.
 D. IV.

☐

II

☐

III

☐

V

☐

IV

☐

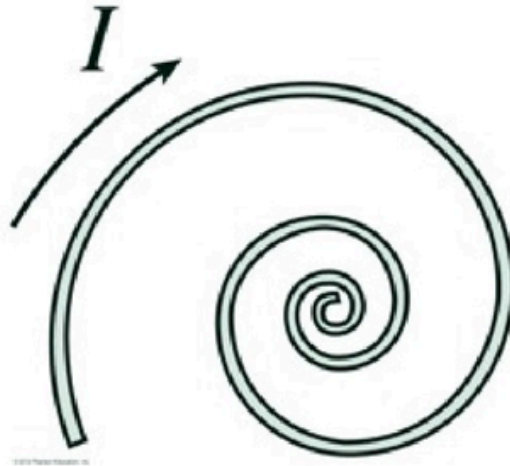
I

⋮

Question 14 5.5 pts

A flexible wire is wound into a flat spiral as shown in the figure. If a current flows in the direction shown, will the coil tighten or loosen?

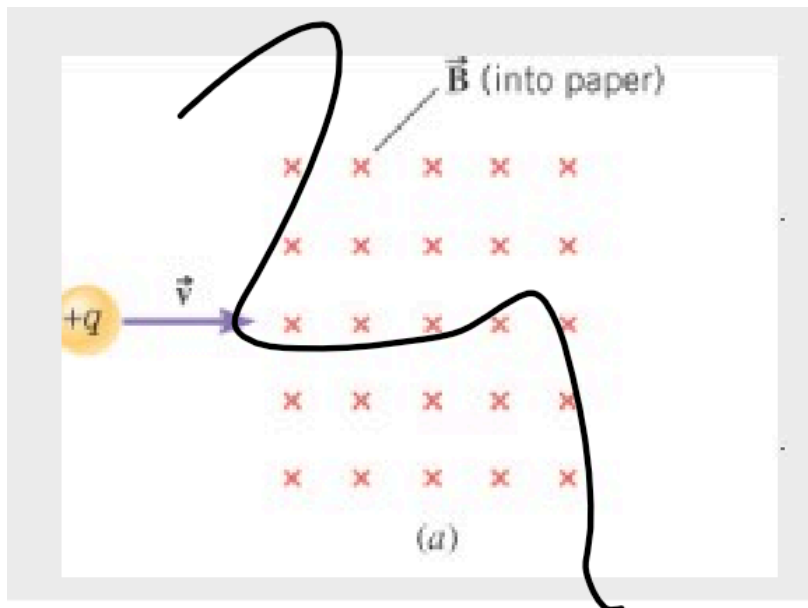
- A. The coil will tighten.
- B. The coil will loosen.



Bonus question

- ☐ B
- ☐ A
- ☒ D

Question 15 5.5 pts



what is true? (don't pay attention to the black line)



the proton is deflected downward



the proton is deflected upward



the proton accelerates

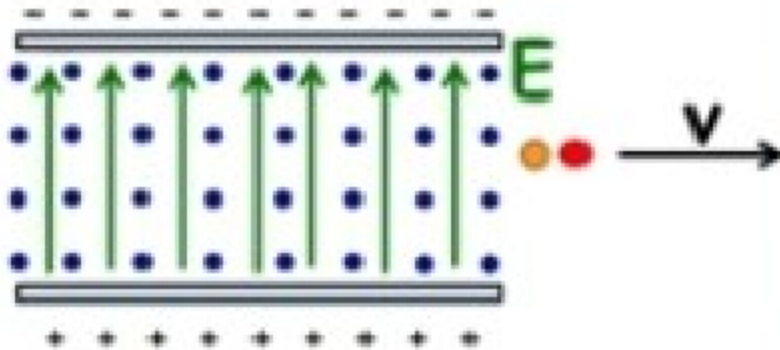


the proton is no deflected



Question 16 5.5 pts

• B (out of page)



positive charged isotopes enter this region with a velocity V . There is an electric field between the parallel plates of electrodes. and there is an electric field as show, What is true



the magnetic force is 0



both the electric field and the magnetic force are up



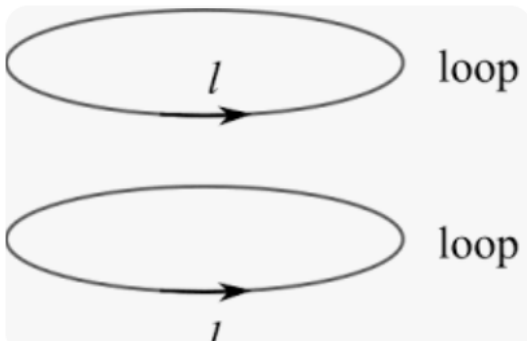
The electric force is up and the magnetic field is down



the electric force is down and the magnetic force is up



Question 17 5.5 pts

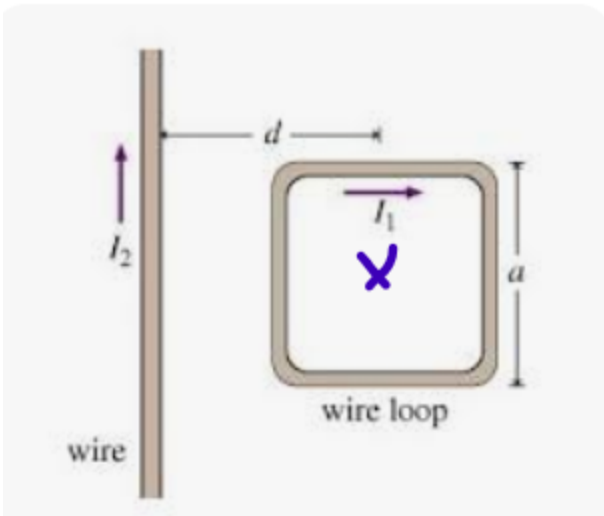


These 2 loops of current

- ☐ repel each other
- ☐ attract each other
- ☐ it depends on the magnitude of the current in each loop
- ☐ have no effect on each other



Question 18 5.5 pts



The cross shows the point C. B2 is the magnetic field is produced by I2. B1 is the magnetic field by I1.

You have a wire and you have a loop. What is true: At the point C

- ☐ B2 is in the screen , B1 is in the screen
- ☐ B2 is in the screen, B1 is out of the screen (toward u)
- ☐ B2 is out of the screen (toward you), B1 is in the screen



B1 and B2 are both out of the screen

Not saved

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